



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Carlos Andres Hernandez
18 July 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

Data Collection

- Collected launch records from the SpaceX REST API and web-scraped historical Falcon 9/Heavy launch data from Wikipedia.

Data Wrangling

- Cleaned the merged dataset, engineered a binary landing-outcome label (“Class”), and loaded it into a SQLite database for querying.

EDA with Visualization & SQL

- Explored how flight number, payload mass, launch site, and orbit relate to landing outcome using Matplotlib/Seaborn and SQL queries.

Interactive Visual Analytics

- Built an interactive Folium map of launch sites/outcomes and a Plotly Dash dashboard for exploring success rate by site and payload.

Predictive Modeling

- Trained Logistic Regression, SVM, Decision Tree, and KNN classifiers, tuned with GridSearchCV, and compared test accuracy.

Summary of Results

Data Insights

Launch site and payload mass are linked to landing outcome; GTO missions show a much lower success rate than LEO, ISS, or VLEO missions.

Model Performance

Logistic Regression, SVM, and KNN each reached 83.33% test accuracy; the Decision Tree classifier reached 94.44%.

Key Findings

The Decision Tree model is the strongest predictor of Falcon 9 first-stage landing success on this dataset.

Reliability Trend

Landing success climbed from 0% before 2013 to over 80% by 2019–2020 as SpaceX refined the booster recovery process.

Introduction

Project Background and Context

- | SpaceX advertises Falcon 9 launches at a fraction of the cost of competing providers, largely because it can recover and reuse the first stage. This project applies the full data science methodology — collection, wrangling, EDA, visualization, and machine learning — to predict whether the first stage will land successfully, which makes it possible to estimate launch cost and gives competitors a benchmark for bidding against SpaceX.

Problems We Want to Find Answers To

- What factors influence the successful landing of the Falcon 9 first stage?
- How can we accurately predict the landing outcome using machine learning models?
- Which machine learning model performs best at predicting landing success?

Section 1

Methodology

Methodology

Executive Summary

- **Data collection methodology:**
- Collected launch records via the SpaceX REST API and web-scraped historical launch data from Wikipedia.
- **Perform data wrangling:**
- Cleaned missing values, engineered the binary landing-outcome label, and standardized features.
- **Perform exploratory data analysis (EDA) using visualization and SQL:**
- Used Matplotlib/Seaborn plots and SQL queries against a SQLite database to explore the data.
- **Perform interactive visual analytics using Folium and Plotly Dash:**
- Built an interactive launch-site map and a launch-records dashboard.
- **Perform predictive analysis using classification models:**
- Built, tuned (GridSearchCV), and evaluated Logistic Regression, SVM, Decision Tree, and KNN models.

Data Collection

- Step 1: Collected launch records from the SpaceX REST API (the /v4/launches endpoint).
- Step 2: Web-scraped historical Falcon 9 and Falcon Heavy launch records from Wikipedia using BeautifulSoup.
- Step 3: Combined and cleaned both sources into a single dataset used throughout the rest of the analysis.

Data Collection – SpaceX API

- Requested the SpaceX v4 /launches endpoint with Python's requests library.
- Parsed the JSON response into a Python dictionary and extracted launch date, site, payload mass, booster, and outcome fields.
- Stored the extracted records in a pandas DataFrame for downstream cleaning.

GitHub URL:

https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

Initiate API Request



Fetch Launch Data



Parse JSON Response



Extract Relevant Fields



Store Data in DataFrame

Data Collection - Scraping

- Fetched the “List of Falcon 9 and Falcon Heavy launches” Wikipedia page with Python's requests library.
- Parsed the HTML with BeautifulSoup and extracted the launch-records table.
- Converted the table into a pandas DataFrame and cleaned/standardized the columns.

GitHub URL:

https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

[1330] Flight No.	Date and time (UTC)	Version, Booster ^[1]	Launch site	Payload ^[2]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[482]	F9 B5  B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,630 kg (34,400 lb) ^[2]	LEO	SpaceX	Success	Success (drone ship)
Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. ^[482]									
79	19 January 2020, 15:30 ^[484]	F9 B5  B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[485] (Dragon C205.1)	12,053 kg (26,570 lb)	Sub-orbital ^[436]	NASA (CTS) ^[427]	Success	No attempt
An atmospheric test of the Dragon 2 abort system after Max Q. The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule; ^[486] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. ^[470] The abort test used the capsule originally intended for the first crewed flight. ^[488] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. ^[300] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.									
80	29 January 2020, 14:07 ^[514]	F9 B5  B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,630 kg (34,400 lb) ^[2]	LEO	SpaceX	Success	Success (drone ship)
Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean. ^[502]									
81	17 February 2020, 15:05 ^[520]	F9 B5  B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,630 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km x 396 km (132 mi x 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship ^[524] due to incorrect wind data. ^[505] This was the first time a flight proven booster failed to land.									
82	7 March 2020, 04:50 ^[526]	F9 B5  B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3 )	1,877 kg (4,359 lb) ^[507]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
Last launch of phase 1 of the CRS contract. Carries Berliozmeo, an ESA platform for hosting external payloads onto ISS. ^[500] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part. ^[508] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.									
83	18 March 2020, 12:16 ^[510]	F9 B5  B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,630 kg (34,400 lb) ^[2]	LEO	SpaceX	Success	Failure (drone ship)
Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019). ^[511] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Meftin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit. ^[512] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor. ^[513]									
84	22 April 2020, 19:30 ^[514]	F9 B5  B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,630 kg (34,400 lb) ^[2]	LEO	SpaceX	Success	Success (drone ship)

Data Wrangling

- Identified and handled missing values, and engineered a binary “Class” landing-outcome label from the Outcome column.
- Launch-site value counts: CCAFS SLC-40: 55, KSC LC-39A: 22, VAFB SLC-4E: 13.
- Orbit value counts: GTO: 27, ISS: 21, VLEO: 14, PO: 9, LEO: 7, SSO: 5, MEO: 3, ES-L1/HEO/SO/GEO: 1 each.
- Exported the cleaned, labeled dataset for use in EDA, SQL analysis, and modeling.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

EDA with Data Visualization

- Flight Number vs. Launch Site, Payload Mass vs. Launch Site — strip plots to see how site usage and payload evolved over time.
- Success Rate by Orbit Type — line chart to compare landing reliability across orbits.
- Flight Number vs. Orbit Type, Payload Mass vs. Orbit Type — scatter plots colored by outcome to spot patterns by orbit.
- Launch Success Rate by Year — line chart to track how reliability improved over time.
- All charts built with Matplotlib/Seaborn on the wrangled dataset.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

EDA with SQL

- Queried the unique launch site names, and records where the site name begins with 'CCA'.
- Calculated total payload mass carried for NASA (CRS) and average payload mass for booster version F9 v1.1.
- Found the date of the first successful ground-pad landing, and boosters with successful drone-ship landings between 4,000–6,000 kg.
- Counted total successful/failure mission outcomes, and identified boosters that carried the maximum payload mass.
- Listed 2015 launch records by month, and ranked landing-outcome counts between 2010-06-04 and 2017-03-20.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

Build an Interactive Map with Folium

- Markers & Circles: placed a circle and labeled marker at each of the 4 launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) to pinpoint their exact locations.
- Color-coded Markers / MarkerCluster: green markers for successful landings, red for failures, clustered per site to compare outcome mix at a glance.
- MousePosition, PolyLine & Distance Marker: measured the distance from a launch site to its nearest coastline using a haversine distance function, and drew a line with the computed distance.
- These objects make it easy to see where SpaceX launches occur, compare success rates by geography, and confirm that launch sites sit right next to the coast for safe over-water flight paths.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

Build a Dashboard with Plotly Dash

- Success Pie Chart: total successful launches by site (All Sites view), or success/failure split for a single selected site.
- Success-Payload Scatter Plot: Payload Mass (kg) vs. launch outcome (class), colored by Booster Version Category.
- Launch Site Dropdown: switch between All Sites and each of the 4 individual launch sites.
- Payload Range Slider: filter both charts to a payload range between 0 and 10,000 kg.
- Together these let a user isolate which sites, payload ranges, and booster versions are associated with the highest landing success.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project/blob/main/7.%20spacex-dash-app.py

Predictive Analysis (Classification)

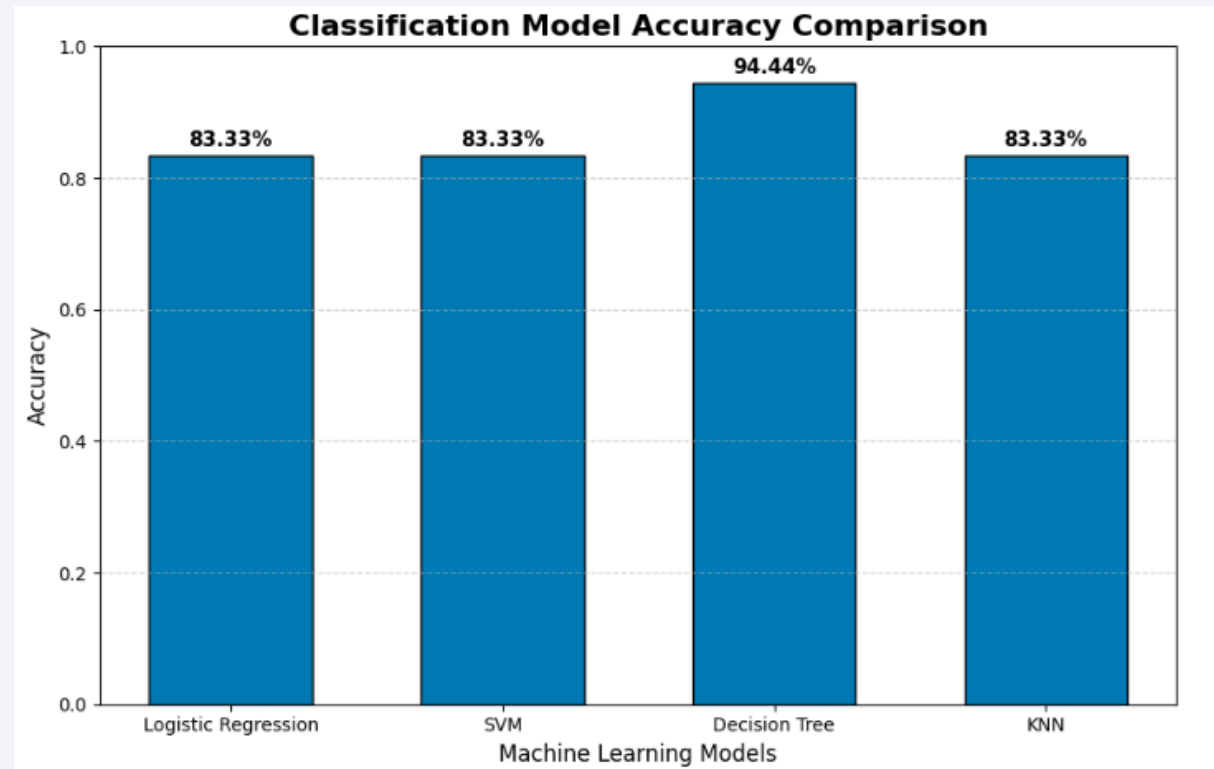
Data Preprocessing → Model Selection → Hyperparameter Tuning (GridSearchCV) → Model Evaluation → Best-Performing Model

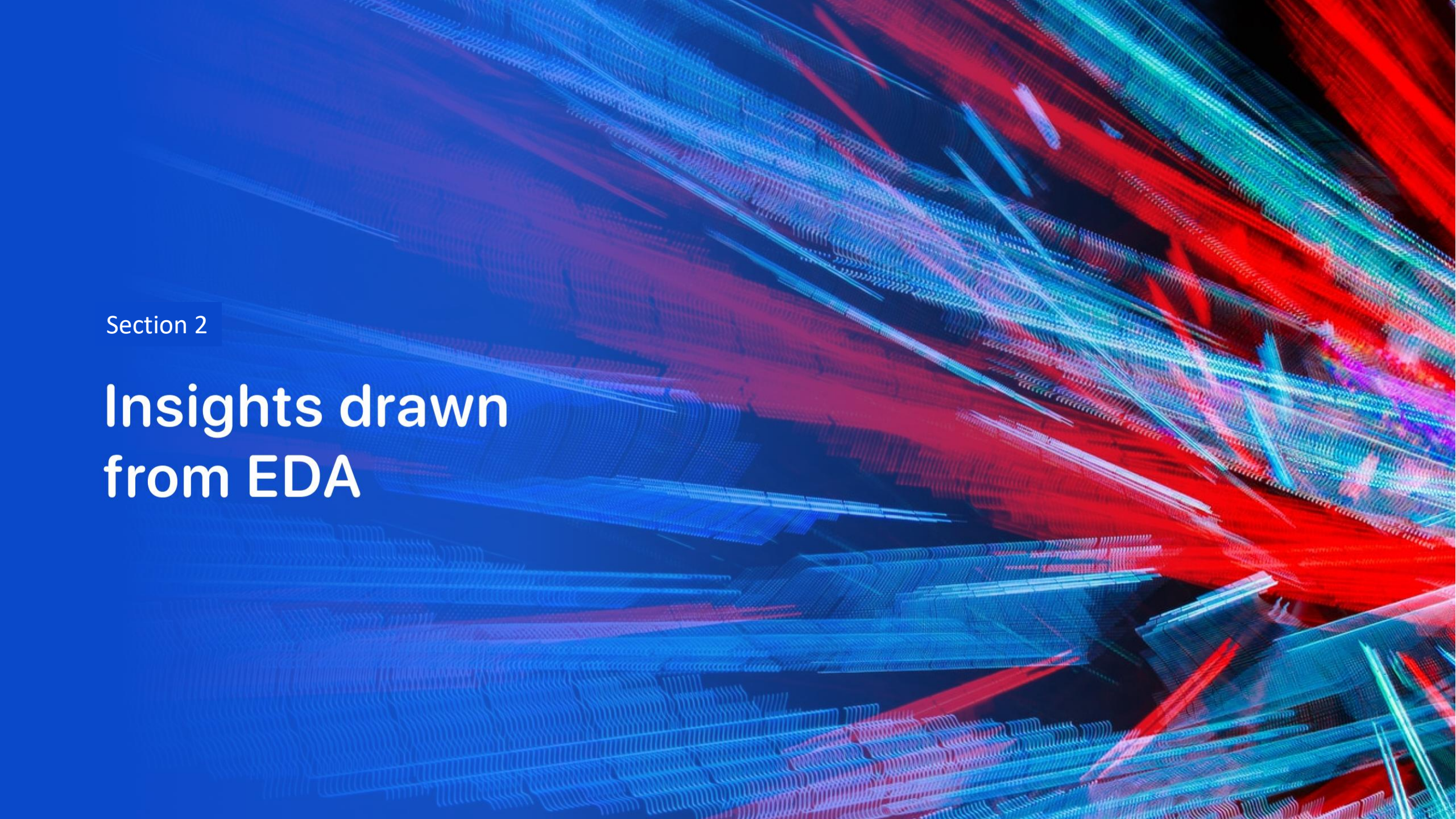
- Standardized features and split the data into train/test sets.
- Trained Logistic Regression, SVM, Decision Tree, and KNN classifiers.
- Tuned each model with GridSearchCV (cv=10) over C/penalty (LR), C/gamma/kernel (SVM), criterion/max_depth (Tree), and n_neighbors/algorithm/p (KNN).
- Evaluated every model on the held-out test set with accuracy and a confusion matrix; the Decision Tree came out on top at 94.44% accuracy.

GitHub URL: https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



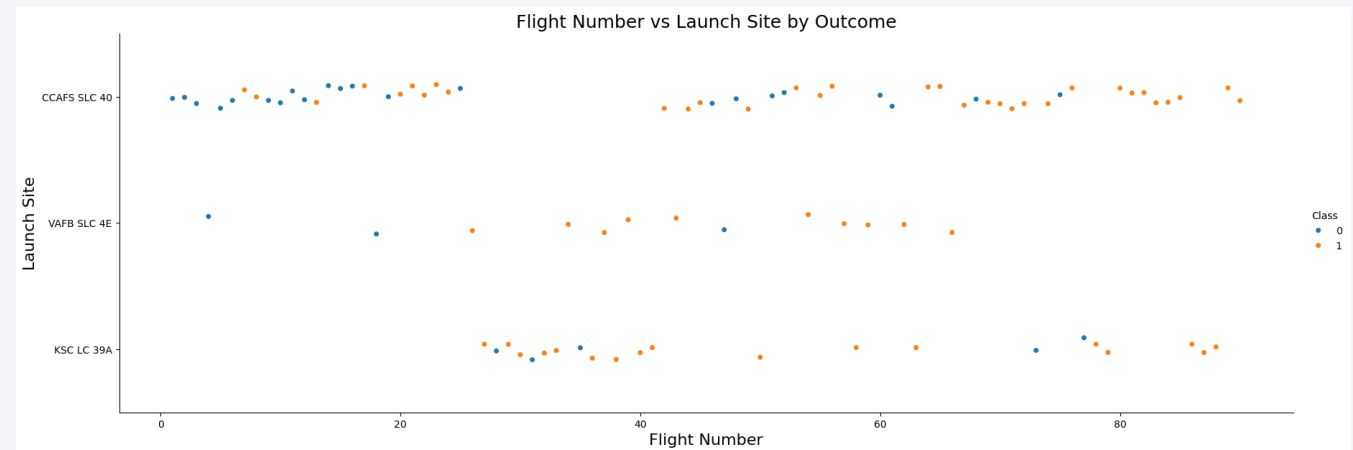
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

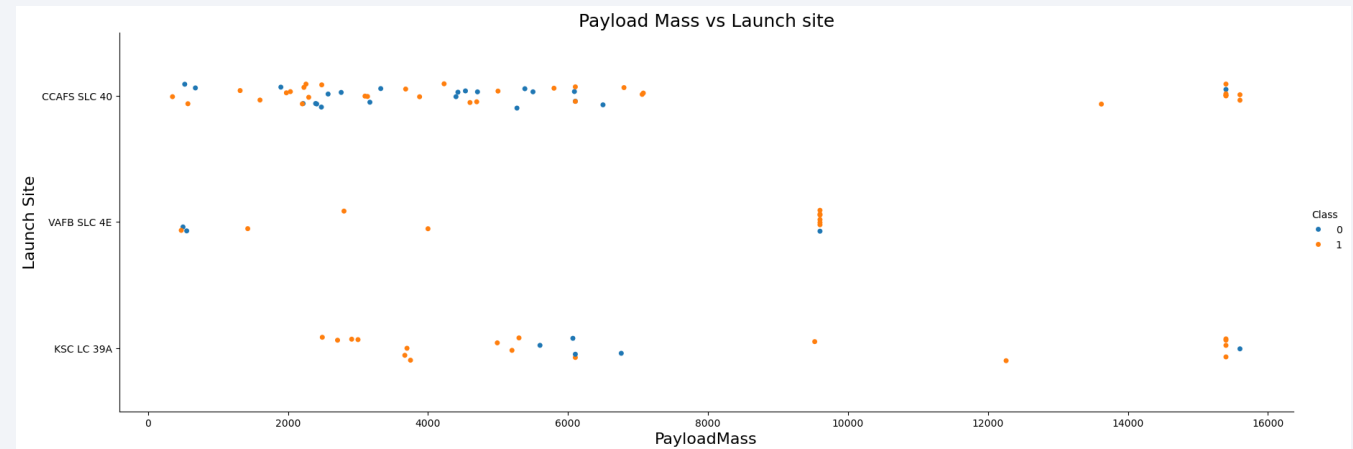
Flight Number vs. Launch Site

- CCAFS SLC 40 has been used across almost the entire flight-number range, with a mix of successful (orange) and unsuccessful (blue) landings.
- KSC LC-39A and VAFB SLC-4E entered service later (higher flight numbers) and show a higher share of successful landings, consistent with growing booster-recovery reliability over time.



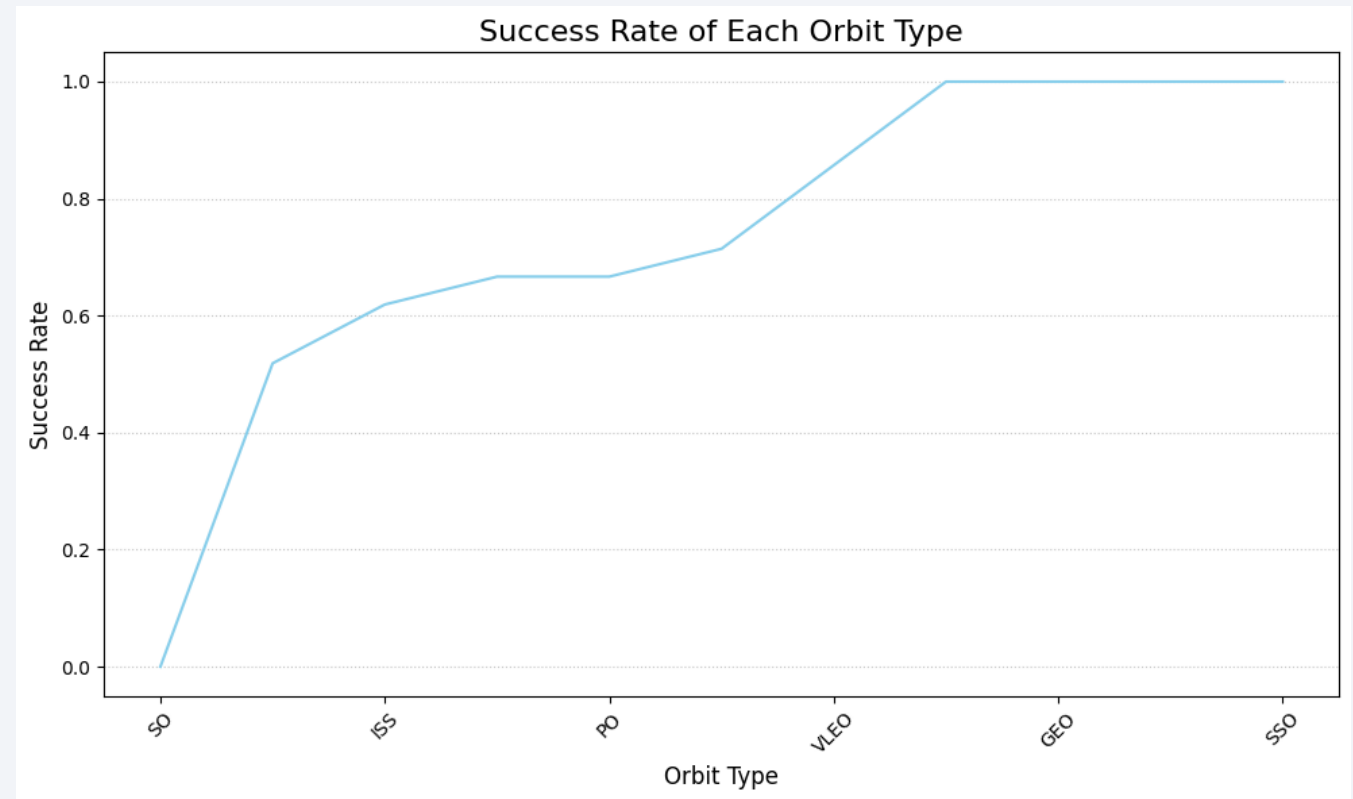
Payload vs. Launch Site

- CCAFS SLC 40 handles the widest spread of payload masses, including the heaviest payloads in the dataset (roughly 13,600–15,600 kg).
- VAFB SLC-4E is not used for payloads much above ~10,000 kg in this dataset, suggesting it is reserved for lighter, polar-orbit missions.



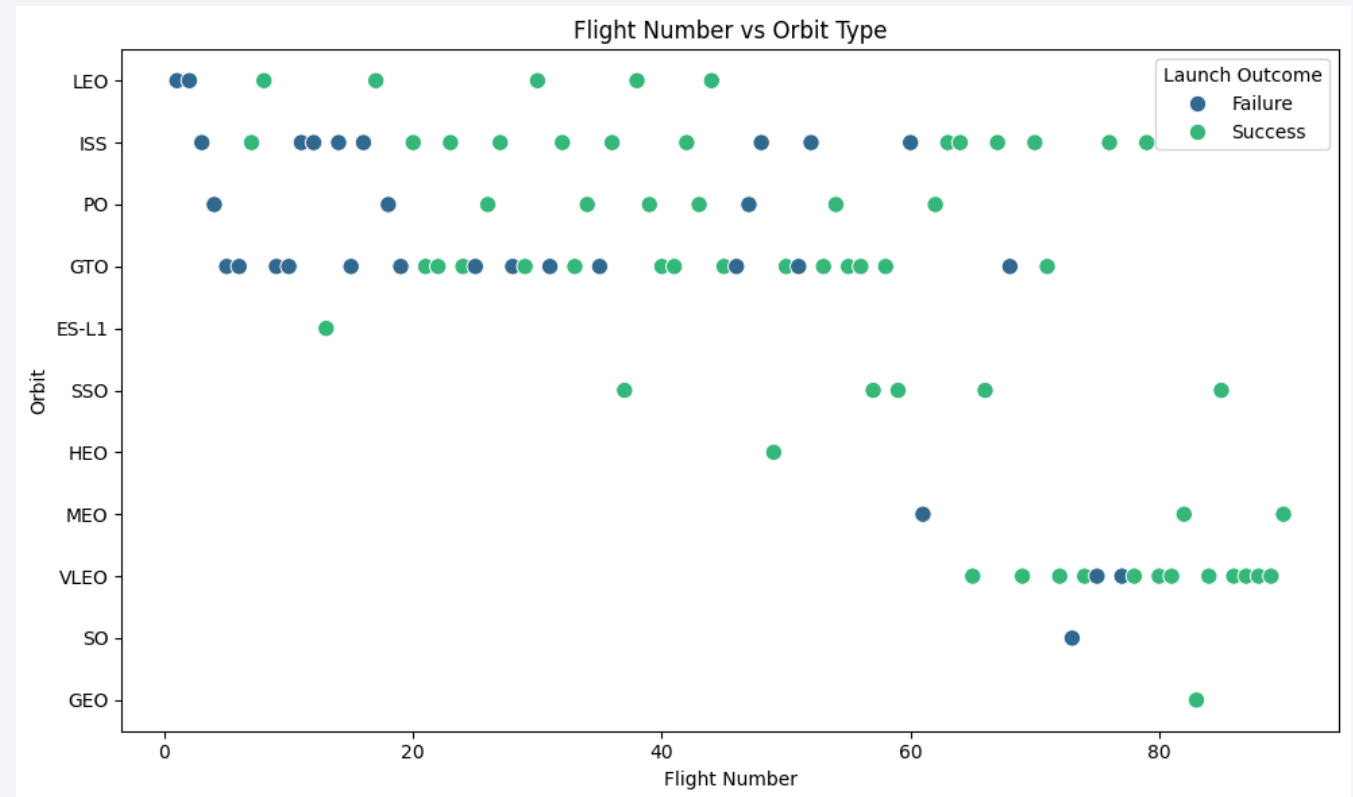
Success Rate vs. Orbit Type

- SO orbit shows a 0% success rate (a single early flight); GTO and ISS sit lowest among frequently-used orbits, in the 50–65% range.
- VLEO, GEO, HEO, ES-L1, and SSO orbits reach close to or exactly 100% success, indicating these missions are the most reliable for first-stage recovery.



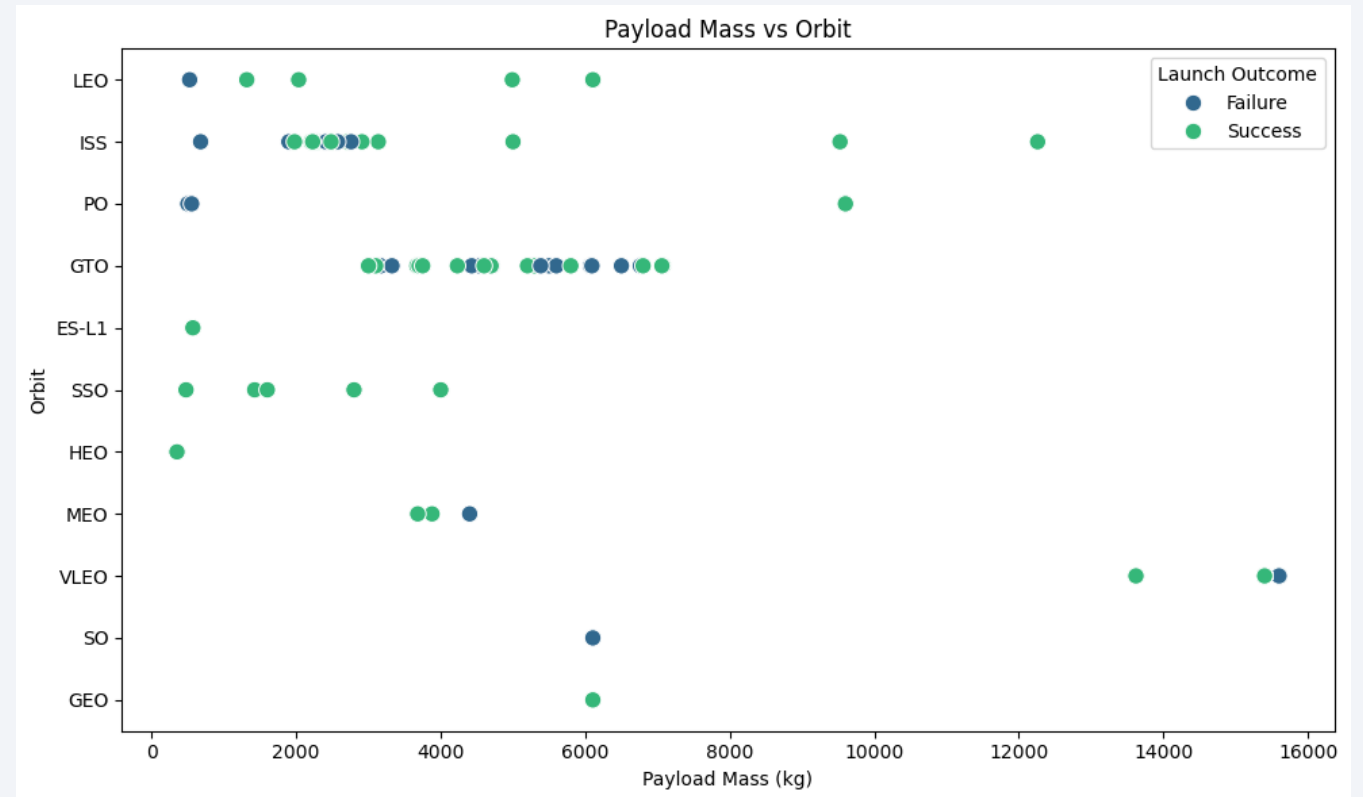
Flight Number vs. Orbit Type

- Early flights (low flight numbers) to GTO and PO orbits include more failures (blue).
- From roughly flight 60 onward, VLEO and ISS missions are almost entirely successful (green), reflecting SpaceX's growing landing reliability.



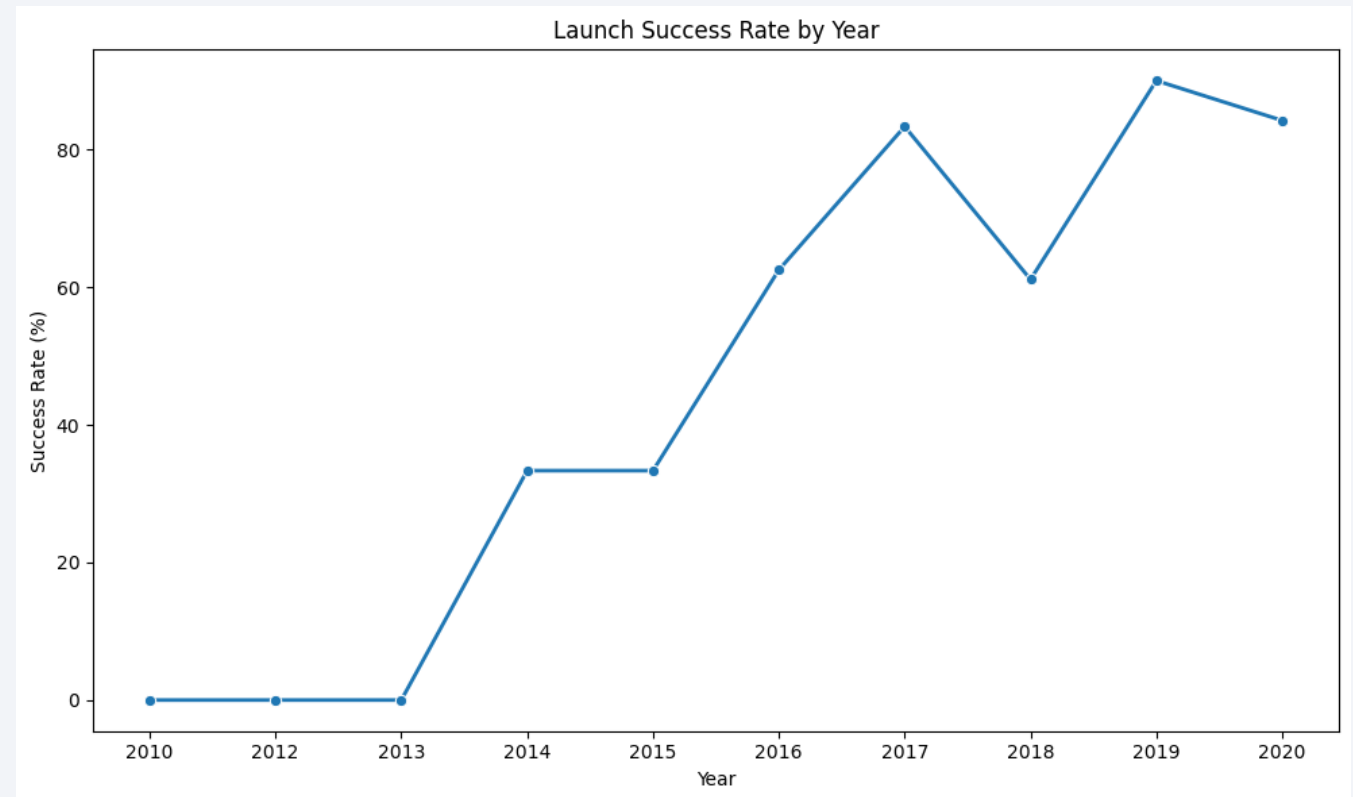
Payload vs. Orbit Type

- The heaviest payloads (>13,000 kg) go to ISS and VLEO orbits and are mostly successful.
- GTO payloads cluster in the 2,000–7,000 kg range and show the most mixed (failure/success) outcomes of any orbit.



Launch Success Yearly Trend

- No successful landings were recorded through 2013.
- Success rate climbs sharply from 2014 (33%) to a peak of 90% in 2019, with a dip to 61% in 2018 before recovering to 84% in 2020.



All Launch Site Names

Query

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Result

CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40

Four distinct launch sites were used for Falcon 9 missions in this dataset.

Launch Site Names Begin with 'CCA'

Query

```
SELECT * FROM SPACEXTABLE  
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Result

5 records returned, all from CCAFS LC-40 – the earliest
Dragon Qualification Unit, demo, and CRS missions (2010–2013).

All of Cape Canaveral's site codes begin with 'CCA', so this filter captures SpaceX's earliest East-Coast missions.

Total Payload Mass

Query

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Result

45,596 kg

Boosters flying NASA Commercial Resupply Services (CRS) missions carried a combined 45,596 kg of payload.

Average Payload Mass by F9 v1.1

Query

```
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

Result

2,928.4 kg

The F9 v1.1 booster family carried an average payload of about 2,928 kg per mission.

First Successful Ground Landing Date

Query

```
SELECT MIN(Date) FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (ground pad)';
```

Result

2015-12-22

SpaceX achieved its first successful ground-pad landing on December 22, 2015 — a milestone for booster reuse.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query

```
SELECT Booster_Version FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (drone ship)'  
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Result

F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Four boosters successfully landed on a drone ship while carrying a mid-weight payload between 4,000 and 6,000 kg.

Total Number of Successful and Failure Mission Outcomes

Query

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE
WHERE Mission_Outcome IN ('Success','Failure')
GROUP BY Mission_Outcome;
```

Result

Success: 98

98 missions are recorded with a successful mission outcome; no rows in this field are labeled simply 'Failure' (mission-level failures are captured separately in the landing-outcome column).

Boosters Carried Maximum Payload

Query

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ =  
      (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Result

12 F9 B5 boosters tied for the maximum payload, incl.
B1048.4/.5, B1049.4/.5/.7, B1051.3/.4/.6, B1056.4,
B1058.3, B1060.2/.3

The heaviest payload in the dataset was carried by 12 different F9 Block 5 boosters, all flying the same maximum payload mass.

2015 Launch Records

Query

```
SELECT CASE WHEN substr(Date,6,2)='01' THEN 'January' ... END AS Month_Name,  
        Mission_Outcome, Booster_Version, Launch_Site  
FROM SPACEXTABLE WHERE substr(Date,0,5) = '2015';
```

Result

Jan-Apr: 5 successes (B1012-B1016)

Jun: 1 failure in flight (B1018)

Dec: 1 success (F9 FT B1019)

2015 had 7 launches, all from CCAFS LC-40 — five early successes, one in-flight failure in June, and the first Full Thrust (FT) booster flight in December.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query

```
SELECT Landing_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC;
```

Result

No attempt: 10 Success (drone ship): 5 Failure (drone ship): 5
Success (ground pad): 3 Controlled (ocean): 3 Uncontrolled (ocean): 2
Failure (parachute): 2 Precluded (drone ship): 1

In this early period, most flights made 'no attempt' at recovery; drone-ship landings were the most common recovery attempt, split evenly between success and failure.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Mark All Launch Sites on a Map

[Insert your Folium map screenshot here — all 4 launch sites marked]

- Are all launch sites close to the Equator? No — VAFB SLC-4E sits at 34.63°N , well north of the three Florida sites, which cluster around $28.5\text{--}28.6^{\circ}\text{N}$.
- Are all launch sites close to the coast? Yes — CCAFS LC-40 ($28.5623, -80.5774$), CCAFS SLC-40 ($28.5632, -80.5768$), and KSC LC-39A ($28.5733, -80.6469$) sit on the Florida coast, and VAFB SLC-4E ($34.6328, -120.6107$) sits on the California coast.

Success/Failure Markers by Launch Site

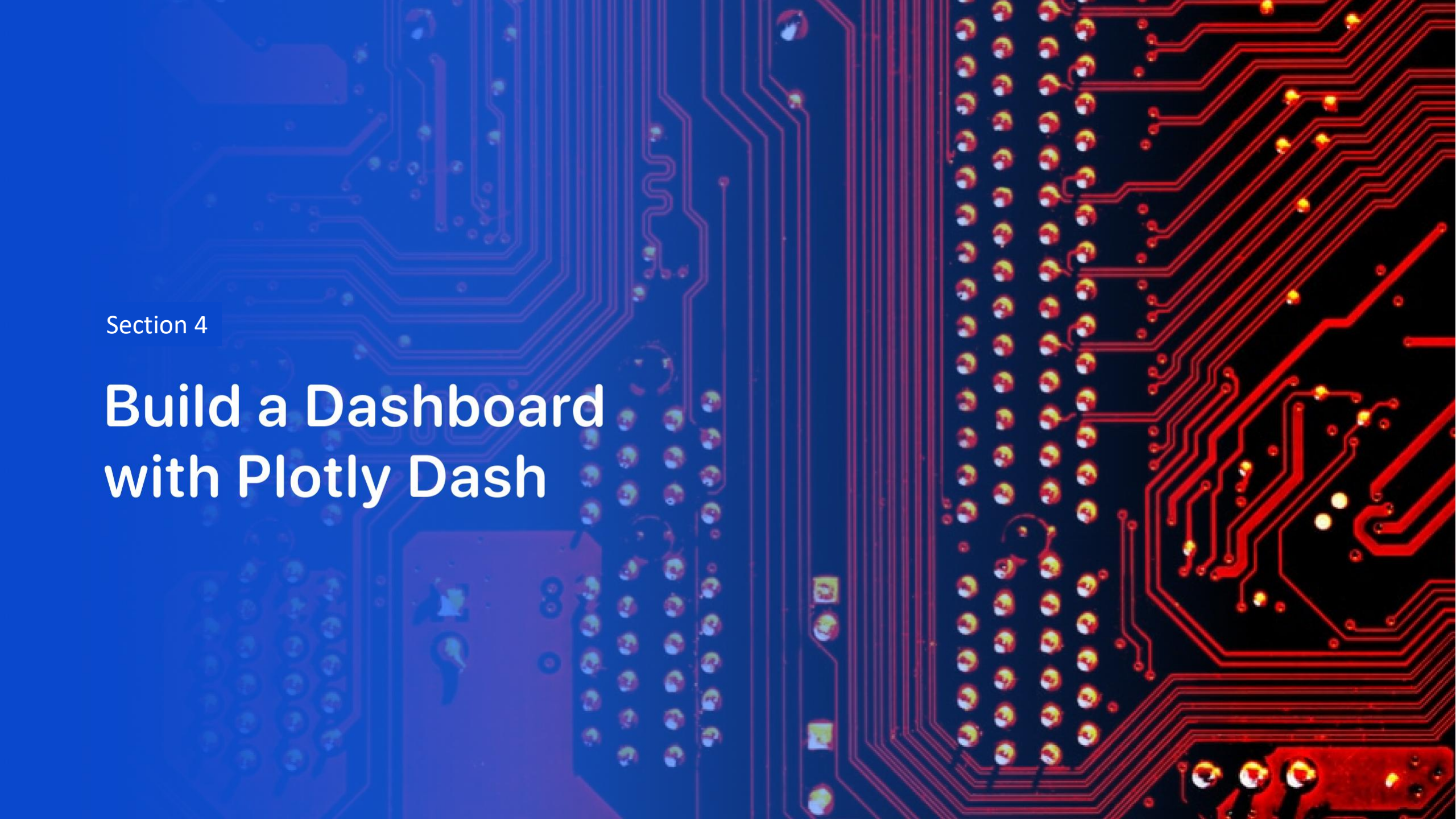
[Insert your Folium screenshot here — color-coded / clustered markers]

- Each launch was plotted as a green (success) or red (failure) marker, clustered by site, so the outcome mix at each site is visible at a glance.
- This view makes it easy to compare sites visually before running the SQL/EDA success-rate numbers — for example, spotting which sites lean more red early in the program.

Launch Site Proximity to the Coastline

[Insert your Folium screenshot here — site-to-coastline distance line]

- Using a haversine distance function, CCAFS SLC-40 is only about 0.58 km from its nearest coastline point (28.56367°N, -80.57163°W).
- This confirms launch sites are sited immediately next to the coast, so ascending rockets fly out over open water and away from populated areas.

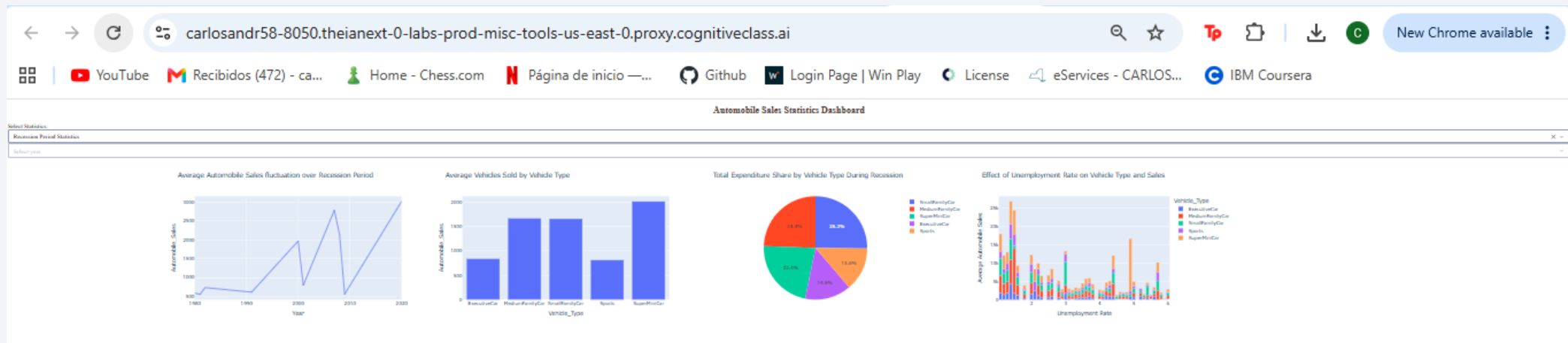


Section 4

Build a Dashboard with Plotly Dash

Launch Success Count for All Sites

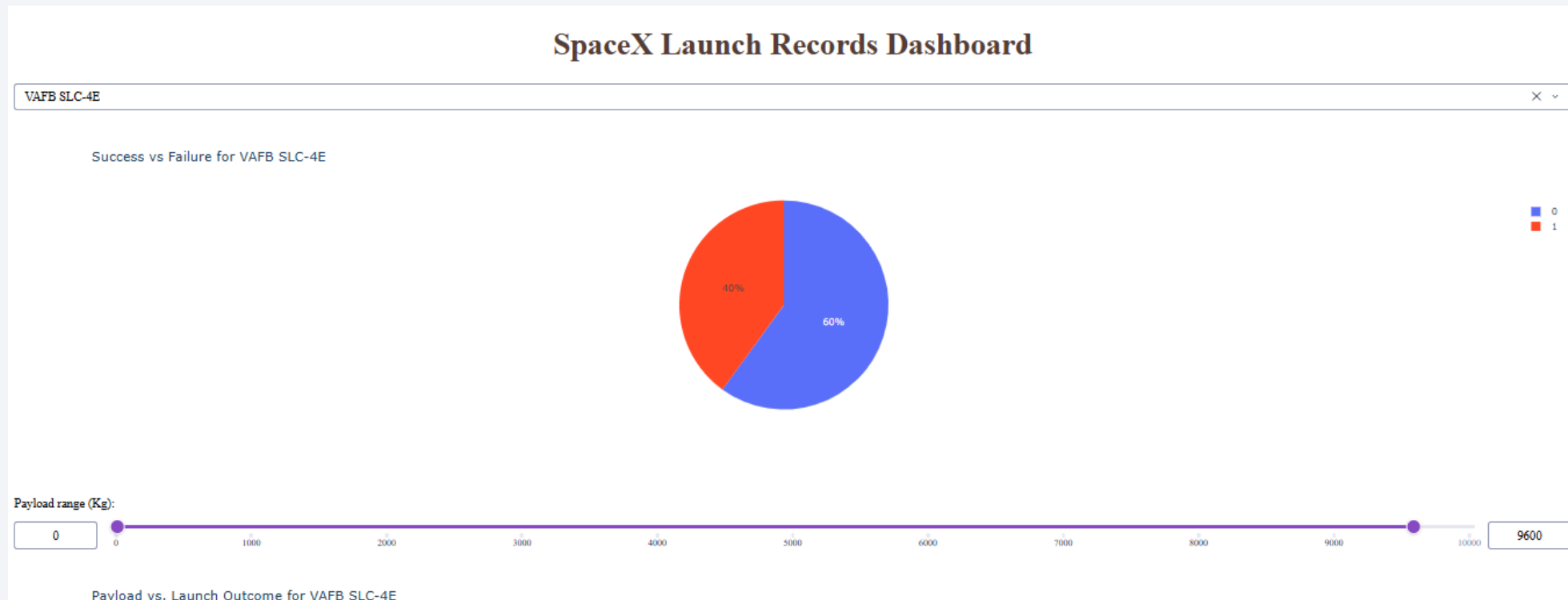
- The pie chart breaks down total successful launches by site: KSC LC-39A leads with 41.7% of successes, followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%).



Success Ratio for the Top Launch Site

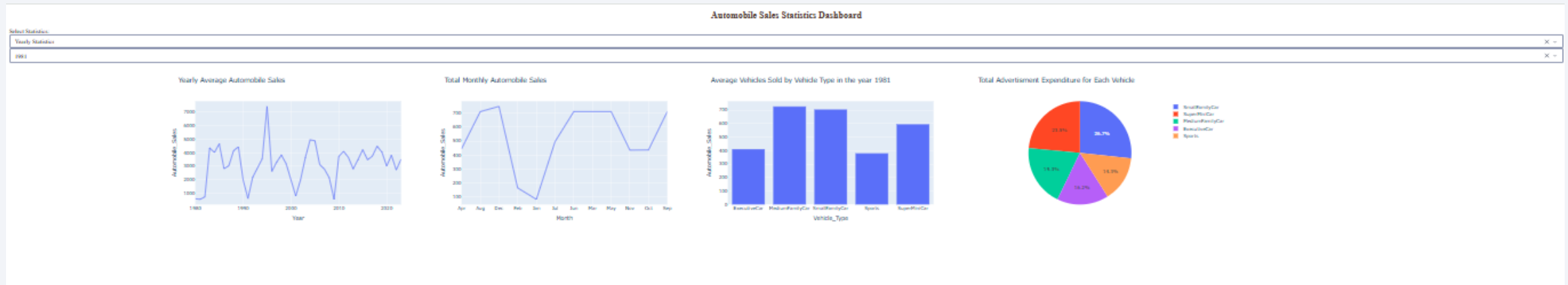
[Insert your dashboard screenshot here — pie chart for KSC LC-39A]

- Selecting KSC LC-39A (the top site) in the dropdown shows a 76.9% success / 23.1% failure split, confirming it as SpaceX's most reliable launch site in this dataset.



Payload vs. Launch Outcome

- Filtering the payload-range slider shows that lighter payloads (under ~2,000 kg) and the FT/B4/B5 booster versions have the highest success rates, while early v1.0/v1.1 boosters cluster with more failures.

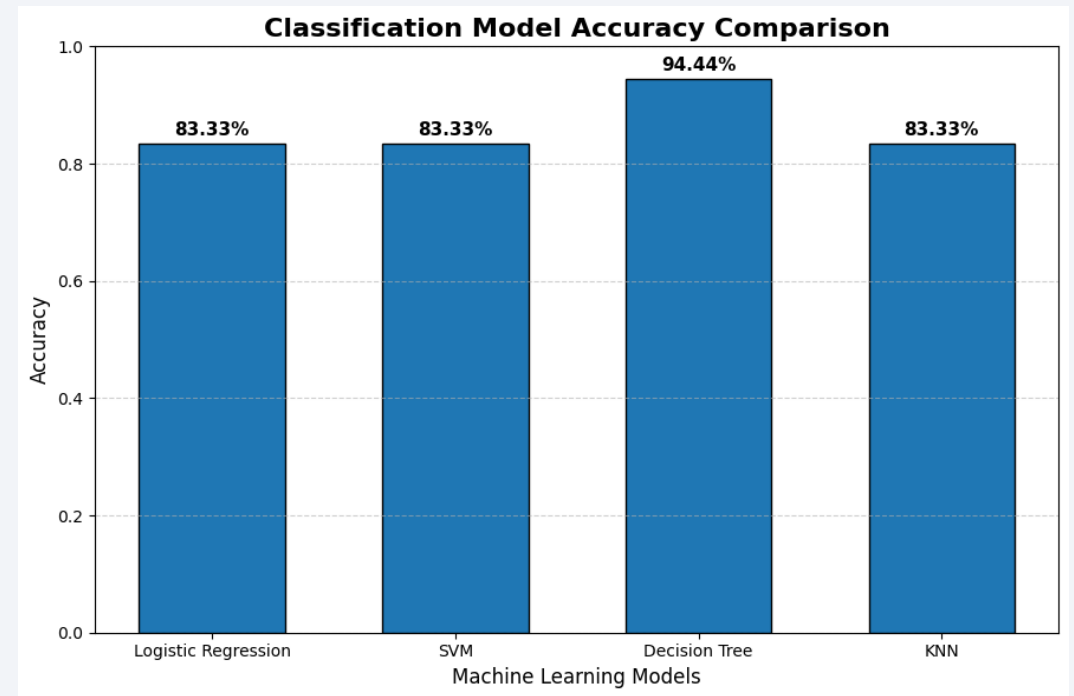


Section 5

Predictive Analysis (Classification)

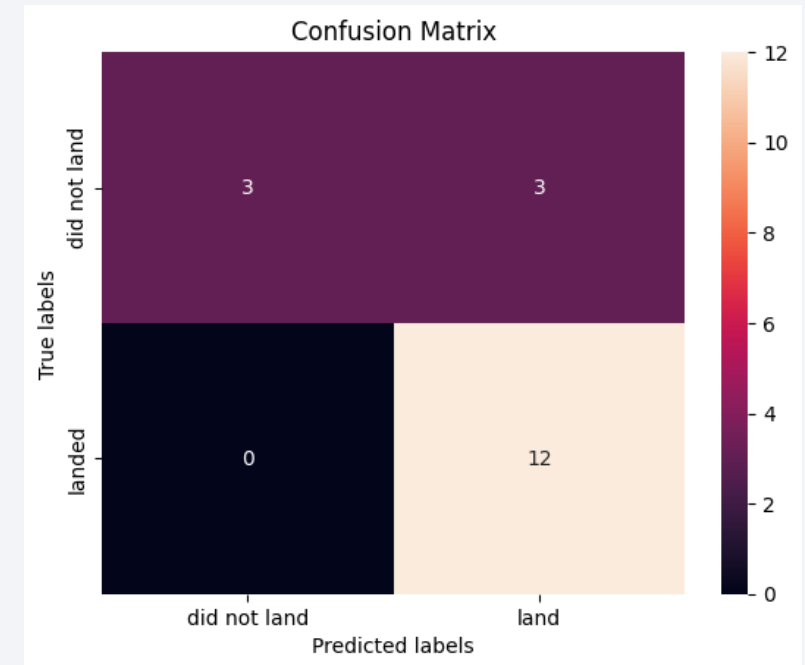
Classification Accuracy

- The Decision Tree model achieved the highest test accuracy at 94.44%.
- Logistic Regression, SVM, and KNN each reached 83.33% accuracy on the held-out test set.



Confusion Matrix

- The Decision Tree (best-performing model) correctly classified 15 of 18 test cases: 5 true “did not land”, 12 true “landed”, 1 false positive, and 0 false negatives.
- Zero false negatives means every actual successful landing was correctly identified — important for cost/readiness planning, where under-predicting success is riskier than over-predicting it.



Conclusions

Point 1:

KSC LC-39A had the highest launch-success ratio of any site (76.9% of its launches succeeded) and the largest share of the program's total successes (41.7%).

Point 2:

Landing reliability improved sharply over time — from 0% before 2013 to roughly 84–90% success in 2019–2020 — as SpaceX iterated on the booster-recovery process.

Point 3:

GTO missions have a noticeably lower success rate than LEO/ISS/VLEO missions, suggesting orbit type (and the payload masses associated with it) affects landing difficulty.

Point 4:

Among the four classifiers tested, the Decision Tree model best predicts first-stage landing success (94.44% test accuracy vs. 83.33% for Logistic Regression, SVM, and KNN).

Appendix

- SQL setup: `%sql sqlite:///my_data1.db — CREATE TABLE SPACEXTABLE AS SELECT * FROM SPACEXTBL WHERE Date IS NOT NULL;`
- Distance helper (Folium notebook): haversine-based `calculate_distance(lat1, lon1, lat2, lon2)` used to measure each launch site's distance to its nearest coastline.
- Model tuning: `GridSearchCV(estimator, param_grid, cv=10)` used to tune Logistic Regression, SVM, Decision Tree, and KNN before comparing test accuracy.
- Notebook outputs referenced throughout this deck are available in the project's GitHub repository:
https://github.com/hercandres/IBM_Applied_Data_Science_Capstone_Project

Thank you!

